



AR VR unit 1 and 2

Augmented and Virtual Reality (Savitribai Phule Pune University)



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COURSE CONTENTS		
Unit I	Introduction to Virtual Reality	(6 hrs)
Defining Virtual Reality, History of VR, Human Physiology and Perception, Key Elements of Virtual Reality Experience, Virtual Reality System, Interface to the Virtual World-Input & output- Visual, Aural & Haptic Displays, Applications of Virtual Reality.		

Defining Virtual Reality

Virtual reality (VR) refers to a computer-generated simulation of a three-dimensional environment or situation that can be interacted with in a seemingly real or physical way using specialized devices such as VR headsets or controllers. The goal of VR is to immerse the user in a simulated experience that feels as if they are actually present in that environment. This technology can be used for a variety of purposes such as entertainment, education, training, and therapy.

- Virtual reality (VR) is a technology that uses computer-generated simulations to create immersive, interactive environments that can be explored in 3D.
- VR typically involves the use of specialized hardware such as head-mounted displays (HMDs), haptic feedback devices, and hand controllers to provide a more realistic and immersive experience.
- VR can be used for a wide range of applications, including entertainment (such as gaming and movies), education and training (such as flight simulation and medical training), and therapy (such as exposure therapy for phobias).
- The goal of VR is to create a sense of presence, or the feeling that the user is actually present in the virtual environment. This is achieved through a combination of sensory inputs, such as visual and auditory cues, as well as haptic feedback and other forms of sensory stimulation.
- VR technology is rapidly advancing, with new developments in areas such as graphics, processing power, and motion tracking enabling more immersive and realistic experiences.
- Some potential benefits of VR include enhanced learning and training, improved accessibility for people with disabilities, and the ability to experience things that may be too dangerous or difficult to experience in real life.
- However, there are also some potential risks associated with VR, such as motion sickness, eye strain, and the potential for addiction or overuse. It is important to use VR technology responsibly and in moderation.

History Of VR:

Virtual reality (VR) has a long and fascinating history, spanning several decades of technological development and experimentation. Here is a brief overview of some key milestones in the history of VR:

- In the 1960s, Ivan Sutherland and his colleagues at MIT developed the first head-mounted display (HMD), a device that could display simple wireframe graphics in 3D. This was a major breakthrough in the development of VR technology.
- In the 1980s, Jaron Lanier and his colleagues founded VPL Research, a company that produced some of the earliest commercial VR products, including HMDs and data gloves for hand tracking.
- In the 1990s, companies such as Sega, Nintendo, and Sony began to release consumer-oriented VR products, such as the Sega VR and the Virtual Boy. However, these early products were often expensive, clunky, and failed to capture mainstream interest.
- In the 2000s, the development of faster processors and more powerful graphics cards allowed for more advanced VR experiences, such as military flight simulators and medical training simulations.
- In 2012, the Oculus Rift, a high-end VR headset designed for gaming, was announced on Kickstarter and quickly became a sensation. This helped to spark renewed interest in VR technology among consumers and developers.
- Since then, there has been a surge of interest and investment in VR, with companies such as Google, Samsung, and HTC releasing their own VR devices, and a growing ecosystem of VR software and applications emerging across various industries.

Today, VR technology continues to evolve rapidly, with ongoing advancements in areas such as graphics, haptic feedback, and motion tracking. As VR becomes more accessible and affordable, it is expected to have a major impact on a wide range of industries, from entertainment and education to healthcare and beyond.

Human Physiology and Perception in Virtual Reality

The human physiology and perception in virtual reality:

- The human visual system plays a crucial role in VR, as it is responsible for processing the visual cues that create the illusion of a 3D environment. The eyes capture images that are sent to the brain, where they are processed and combined to create a sense of depth and perspective.
- In VR, the visual cues are provided by the computer-generated environment displayed on the headset or screen. This can create the illusion of a fully immersive 3D environment that surrounds the user.

- The human auditory system also plays a role in VR, as realistic sound effects and spatial audio can enhance the sense of presence and immersion in the virtual environment.
- The sense of touch and proprioception (the awareness of one's own body position and movement) are also important in VR, and can be provided through haptic feedback devices, such as gloves or vests that provide vibrations or other tactile sensations.
- In terms of physiology, VR can have both positive and negative effects on the human body. On the positive side, VR can provide a sense of presence and immersion that can enhance learning and training, and can even provide therapeutic benefits for certain conditions such as phobias and PTSD.
- However, VR can also cause negative effects such as motion sickness, eye strain, and headaches, particularly if the virtual environment does not match up with the user's real-world movements and sensations.
- To mitigate these negative effects, VR designers and developers must carefully consider factors such as frame rate, latency, and field of view, and ensure that the virtual environment is as realistic and comfortable as possible.
- Overall, the human physiology and perception play a crucial role in the success and effectiveness of VR technology, and continued research and development in these areas will be key to unlocking its full potential.

Key Elements of VR Experience:

There are several key elements that contribute to a successful virtual reality (VR) experience:

Virtual World:

- The virtual world is the environment created by the VR system, which can be designed to resemble the real world or a completely fictional world.
- The virtual world should be visually compelling, with realistic graphics and details that enhance the user's sense of presence and immersion.
- The virtual world should be interactive, allowing the user to explore and interact with objects and other elements within the environment.
- The virtual world should have a sense of scale and depth, giving the user a sense of physical presence within the environment.

Immersion:

- Immersion is the sense of being fully absorbed and engaged in the VR experience, feeling like you are truly present in the virtual environment.

- Immersion can be enhanced by realistic graphics, sound, and other sensory feedback that make the virtual world feel believable and compelling.
- Immersion can also be enhanced by minimizing distractions or interruptions that might pull the user out of the virtual world, such as notifications or technical issues.
- In VR, immersion is physical and mental.
- **Immersion** is a sensation of being in an environment; can be a purely mental state or can be accomplished through physical means.
- **Mental immersion** is a state of being deeply engaged; suspension of disbelief; involvement.
- **Physical immersion** is a bodily entering into a medium; synthetic stimulus of the body's senses via the use of technology.
- Physical immersion is a defining characteristic of virtual reality while mental immersion is probably the goal of most media creators.
- **Presence**: sense of presence; being mentally immersed.

Sensory Feedback:

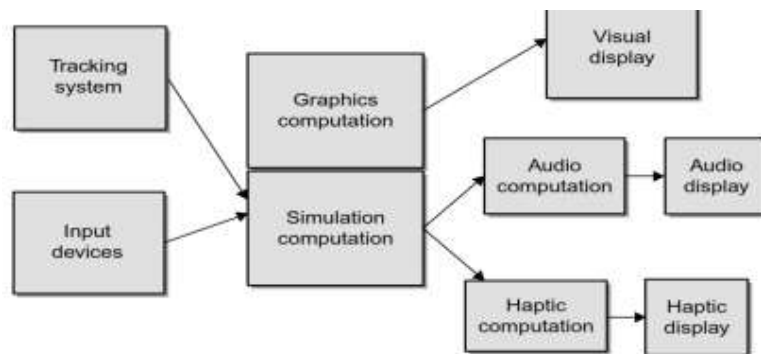
- Sensory feedback is the use of audio, visual, and haptic displays to provide feedback to the user's senses, enhancing the sense of presence and immersion within the virtual world.
- Sensory feedback can include realistic sound effects, directional audio cues, and spatial sound to create a convincing audio environment.
- Sensory feedback can also include realistic graphics, lighting, and textures to create a visually compelling environment.
- Haptic displays can provide touch feedback, force feedback, vibration, and other physical sensations to enhance the user's sense of physical presence within the virtual world.
- The VR system provides direct sensory feedback to the participants based on their physical position.
- A typical VR system will track the head of the participant and at least one hand or an object held by the hand to generate sensory output.
- **Position tracking** is the computerized sensing of the position (location and/or orientation) of an object in the physical world-usually at least part of the participant's body.

Interactivity:

- Interactivity refers to the ability of the user to interact with objects and other elements within the virtual environment.
- Interactivity can be enhanced by providing natural and intuitive controls, such as hand gestures or physical movements, to interact with objects in the virtual world.
- Interactivity can also be enhanced by providing realistic physics and environmental interactions, allowing objects to behave in a believable and predictable way.
- Multiplayer functionality can also enhance interactivity, allowing users to interact with each other within the virtual environment.

Virtual Reality System:

A virtual reality (VR) system is a combination of hardware and software that is designed to create a fully immersive virtual environment for the user. At its core, a VR system typically consists of the following components:



from PPT

1. **Display:** A VR headset or display is used to provide the user with a virtual visual environment. The display should provide a wide field of view and high resolution graphics to create a realistic and immersive experience.
2. **Input devices:** A variety of input devices can be used in a VR system, including hand controllers, tracking sensors, and cameras. These devices allow the user to interact with the virtual environment, and can provide haptic feedback to simulate touch and movement.
3. **Computer hardware:** A powerful computer is typically required to run VR software and provide the high-quality graphics and processing power needed for a smooth and responsive VR experience.

Hardware: Computer/Graphics Engine

- Responsible for calculating the physical behavior of the virtual world,
 - Renders the state of the world into visual, aural, haptic, etc. Representations.
 - Must require
 - Enough computational power
 - Ability to synchronize the display updates between multiple displays
 - Ability to render sounds
 - Ability to perform multiple operations at the same time
 - Can be
 - a single large computer that meets all the requirement
 - interconnected via a low-latency, high-speed communication network
4. Software: VR software is used to create and control the virtual environment, including graphics, sound, and interactivity. This software may be built on a gaming engine or other programming platform.
 5. Tracking system: To ensure accurate and responsive movement tracking, a VR system typically includes a tracking system that can monitor the user's movements and adjust the virtual environment accordingly.
 6. Audio system: Spatial audio is an important component of a VR experience, and a high-quality audio system is needed to create realistic sound effects and directional audio cues.

Together, these components work to create a fully immersive and interactive virtual environment for the user, with realistic visuals, sound, and haptic feedback that simulate a real-world experience. As VR technology continues to evolve, these components are likely to become more sophisticated and advanced, leading to even more immersive and engaging VR experiences.

Interface to the Virtual World input and output - visual

The visual interface to the virtual world in a virtual reality (VR) system typically involves a display or headset that provides a high-quality visual experience for the user. This display or headset is designed to fully immerse the user in a virtual environment, creating a sense of presence and allowing the user to interact with the environment in a natural and intuitive way.

There are several key features that are important for a high-quality visual interface in a VR system. These include:

1. **High resolution:** The display or headset should provide a high-resolution image that is clear and sharp, with no visible pixels or distortion.
2. **Wide field of view:** A wide field of view is important for creating a sense of immersion and presence, as it allows the user to see more of the virtual environment and feel as though they are truly present in it.
3. **Low latency:** The display or headset should have low latency, meaning that there is minimal delay between the user's movements and the display of the virtual environment. This is important for creating a smooth and responsive experience that feels natural and intuitive.
4. **Realistic colors and lighting:** The virtual environment should be rendered with realistic colors and lighting, to create a convincing and immersive experience for the user.
5. **Stereoscopic vision:** Many VR systems use stereoscopic vision to provide a sense of depth and dimensionality, allowing the user to perceive the virtual environment as three-dimensional and interact with it in a more natural and intuitive way.

Overall, the visual interface to the virtual world in a VR system is a crucial component that helps to create a fully immersive and engaging experience for the user. By providing high-quality visuals that are realistic and responsive, a VR system can create a sense of presence that feels like a true escape from the real world.

Explain aural and haptic displays:

Aural Displays:

Aural displays are a type of interface that use sound to convey information or data to a user. Aural displays can be found in a variety of contexts, such as in cars, airplanes, and medical devices.

One of the advantages of aural displays is that they can be used to convey information in situations where visual displays may not be practical or safe, such as when a driver needs to keep their eyes on the road. Aural displays can also be useful for people with visual impairments who may not be able to use visual displays.

Aural displays can take many forms, including simple beeps and chimes, as well as more complex sound patterns and spoken messages. Designing effective aural displays requires careful consideration of factors such as the user's environment, the type of information being conveyed, and the user's familiarity with the sounds being used.

Overall, aural displays can be a valuable tool for conveying information to users in a variety of situations where visual displays may not be practical or safe.

1. **Spatial sound:** Aural displays can include spatial sound, which creates a sense of presence by allowing the user to perceive sound as coming from specific locations within the virtual environment.
2. **Directional audio cues:** Aural displays can also include directional audio cues, which provide feedback to the user about their position and orientation within the environment. For example, the user may hear footsteps approaching from behind.
3. **Immersion:** Aural displays can significantly enhance the immersive experience of a VR system by providing realistic spatial sound and directional audio cues.
4. **Realism:** Aural displays can create a more realistic and engaging environment by providing audio cues that mimic real-world sounds and behaviors.
5. **Interactivity:** Aural displays can enhance the interactivity of a VR system by allowing the user to interact with the environment through sound, such as by communicating with other users or triggering sound effects.
6. **Accessibility:** Aural displays can make VR systems more accessible for users who are visually impaired, by providing valuable feedback through sound.
7. **Personalization:** Aural displays can be customized to the preferences of the user, such as by adjusting the volume or balance of sound channels.
8. **Integration:** Aural displays can be integrated with other sensory displays, such as haptic displays, to create a more cohesive and immersive VR experience.

Haptic Displays:

Haptic displays are a type of interface that use touch or tactile feedback to convey information or data to a user. Haptic displays can be found in a variety of devices, such as smartphones, game controllers, and medical devices.

One of the advantages of haptic displays is that they can provide a more immersive and interactive experience for the user. For example, a smartphone can use haptic feedback to simulate the sensation of pressing a physical button, even though the button is not actually present.

Haptic displays can take many forms, including vibrations, pressure, temperature, and texture. Designing effective haptic displays requires careful consideration of factors such as the user's sensitivity to touch, the type of information being conveyed, and the desired level of interaction.

Overall, haptic displays can be a valuable tool for conveying information to users in a more immersive and interactive way. They can enhance the user experience and provide a more intuitive and natural interface for interacting with devices.

1. Touch feedback: Haptic displays can simulate the sensation of touch, providing feedback to the user when they interact with objects in the virtual environment.
2. Force feedback: Haptic displays can also provide force feedback, simulating the sensation of weight or resistance when the user interacts with objects.
3. Vibration: Haptic displays can use vibration to simulate a variety of sensations, such as impact, texture, or motion.
4. Motion tracking: Haptic displays can track the user's movements and provide feedback based on their actions, such as by simulating the sensation of swinging a virtual object.
5. Realism: Haptic displays can create a more realistic and engaging environment by providing physical feedback that mimics real-world sensations.
6. Immersion: Haptic displays can significantly enhance the immersive experience of a VR system by providing realistic physical feedback that makes the user feel like they are truly present in the virtual environment.
7. Accessibility: Haptic displays can make VR systems more accessible for users with limited mobility, by allowing them to interact with the environment through touch and other physical sensations.
8. Interactivity: Haptic displays can enhance the interactivity of a VR system by providing physical feedback that responds to the user's actions, allowing them to interact with the environment in a more natural and intuitive way.

Virtual Reality Applications:

Virtual reality (VR) technology has a wide range of applications across various industries. Here are some common applications of virtual reality:

1. Gaming and entertainment: VR is commonly used in the gaming and entertainment industry, providing immersive experiences for players and users.
2. Training and education: VR is used for training simulations in fields such as medicine, military, aviation, and engineering. It is also used in education, providing interactive and engaging learning experiences.
3. Healthcare and therapy: VR is used in healthcare for pain management, physical therapy, and psychological therapy. It is also used for medical training and education.
4. Architecture and design: VR is used in architecture and design to create virtual models and walkthroughs of buildings and structures, allowing designers to visualize and test designs before construction.
5. Tourism and hospitality: VR is used in tourism and hospitality to provide virtual tours and experiences of destinations and accommodations.

6. Advertising and marketing: VR is used in advertising and marketing to create immersive and engaging brand experiences for customers.
7. Sports and fitness: VR is used in sports and fitness for training, rehabilitation, and virtual competitions.
8. Art and creativity: VR is used in art and creativity for immersive installations and experiences, allowing artists to create interactive and engaging art pieces.
9. Real estate: VR is used in real estate to provide virtual tours and experiences of properties for buyers and investors.
10. Social and communication: VR is used for social and communication purposes, providing virtual social environments and meetings for remote teams and individuals.

Unit II	Representing the Virtual World in VR	(6 hrs)
Representation of the Virtual World, Visual Representation in VR, Aural Representation in VR and Haptic Representation in VR, Geometric Models, Changing Position and Orientation, Axis-Angle Representations of Rotation, Viewing Transformations		

Representation of Virtual World:

Representation of virtual worlds refers to the ways in which digital environments are designed, visualized, and communicated to users. This involves the use of various media and technologies such as computer graphics, 3D modeling, animation, sound, and text to create immersive experiences that simulate the real world or fictional environments.

In virtual worlds, users can interact with digital objects, characters, and other users in real-time, often through avatars or virtual representations of themselves. The representation of virtual worlds is crucial to creating an engaging and immersive experience that feels realistic and believable.

To achieve this, designers and developers use a variety of techniques to create and render virtual environments, including photorealistic rendering, stylized graphics, and immersive soundscapes. They may also use narrative elements, such as storylines, quests, and characters, to give users a sense of purpose and direction within the virtual world.

Overall, the representation of virtual worlds is a complex process that requires careful consideration of various factors such as user experience, aesthetics, functionality, and technical feasibility. As virtual worlds continue to evolve and become more sophisticated, the representation of these environments will likely continue to play a critical role in shaping how users interact with and experience digital content.

Key points:

1. Representation of virtual worlds involves designing, visualizing, and communicating digital environments to users.
2. It requires the use of various media and technologies such as computer graphics, 3D modeling, animation, sound, and text to create immersive experiences.
3. Users interact with digital objects, characters, and other users in real-time, often through avatars or virtual representations of themselves.
4. Designers and developers use techniques like photorealistic rendering, stylized graphics, immersive soundscapes, and narrative elements to create and render virtual environments.
5. The representation of virtual worlds is crucial to creating an engaging and immersive experience that feels realistic and believable.
6. It requires consideration of various factors such as user experience, aesthetics, functionality, and technical feasibility.

7. As virtual worlds continue to evolve and become more sophisticated, the representation of these environments will likely continue to play a critical role in shaping how users interact with and experience digital content.

Visual Representation in VR

Visual representation in VR refers to the way that virtual reality environments are created to look and feel realistic to users. In VR, the user is fully immersed in a digital environment, and the visual representation is a crucial aspect of creating that sense of immersion.

There are several key factors to consider when creating a visually compelling VR experience. First, the graphics need to be of high quality, with realistic textures, lighting, and shadows. This is typically achieved through the use of advanced rendering techniques, such as ray tracing or real-time lighting.

Another important aspect of visual representation in VR is the use of 3D modeling and animation. VR environments are typically created using 3D models, which can be animated to simulate movement and interaction. This allows users to feel like they are really in the virtual environment, rather than just observing it from a distance.

Finally, sound and other sensory feedback can also be used to enhance the visual representation in VR. For example, spatial audio can be used to create a sense of depth and directionality, while haptic feedback can be used to simulate touch and other physical sensations.

Overall, creating a visually compelling VR experience requires a combination of technical expertise, artistic talent, and an understanding of how users perceive and interact with virtual environments. With the right approach, VR can be a powerful tool for creating immersive, engaging experiences that feel almost like real life.

Aural Representation in VR:

Aural representation in VR refers to the use of sound and audio to create an immersive experience within a virtual environment. This involves creating a soundscape that matches the visual elements of the virtual world, as well as the user's movements and interactions within that world.

Aural representation can greatly enhance the user's sense of presence within the virtual environment, making it feel more realistic and immersive. For example, spatial audio can be used to create the illusion of sounds coming from different directions and distances, just as they would in the real world. This can help users to locate and identify objects, characters, or events within the virtual world.

In addition to spatial audio, designers and developers can also use music, ambient sound effects, and other audio elements to create a particular mood or atmosphere within the

virtual environment. This can be especially effective in games, simulations, and other applications that rely on a particular emotional or narrative tone.

Overall, aural representation is a crucial element in creating a truly immersive virtual experience. It can help to enhance the user's sense of presence and engagement with the virtual environment, and can be used to create a wide range of emotional and narrative effects.

Examples of aural representation in VR:

1. **Spatial audio:** This is perhaps the most common form of aural representation in VR. It involves using sound to create the illusion of objects, events, or characters being located at different positions and distances within the virtual environment. For example, if a user is walking through a forest in a VR game, the sounds of birds chirping and leaves rustling might be heard from different directions, creating a more realistic and immersive experience.
2. **Music:** Music can be used to set the tone and mood of a virtual environment. For example, a VR horror game might use eerie music to create a sense of tension and fear, while a VR adventure game might use upbeat and adventurous music to create a sense of excitement and wonder.
3. **Voice acting:** Voice acting can be used to bring characters to life within a virtual environment. For example, in a VR storytelling experience, voice actors might be used to create a more engaging and immersive narrative.
4. **Ambient sound effects:** Ambient sound effects such as wind, water, or traffic sounds can be used to create a more realistic and immersive environment. For example, in a VR cityscape, the sounds of traffic, sirens, and people talking can help to create a sense of being in a bustling urban environment.

Overall, aural representation in VR is a crucial element in creating a truly immersive experience that engages multiple senses and creates a more convincing illusion of reality.

Haptic representation in VR

Haptic representation in VR refers to the use of touch-based feedback to create a more immersive and realistic virtual environment. This involves using haptic devices such as gloves, vests, or controllers that simulate the sensation of touch, pressure, or vibration.

Haptic representation can greatly enhance the user's sense of presence within the virtual environment by providing a more tangible and visceral experience. For example, haptic gloves can be used to simulate the sensation of touching a virtual object, while a haptic vest can provide the sensation of being hit or pushed within the virtual world.

There are various types of haptic feedback that can be used in VR, including:

1. **Tactile feedback:** This involves creating the sensation of touch or pressure on the user's skin. For example, a haptic glove might simulate the sensation of picking up a virtual object by creating a sensation of pressure on the user's fingertips.
2. **Force feedback:** This involves creating the sensation of resistance or force on the user's body. For example, a haptic vest might create a sensation of impact or pressure when the user's virtual avatar collides with an object.
3. **Vibration feedback:** This involves creating a sensation of vibration or shaking on the user's body. For example, a haptic controller might vibrate to simulate the sensation of firing a virtual gun or hitting an object within the virtual world.

Overall, haptic representation in VR is a crucial element in creating a truly immersive and engaging experience. It can greatly enhance the user's sense of presence and interaction with the virtual environment, and can be used to create a wide range of sensory experiences that feel more realistic and tangible.

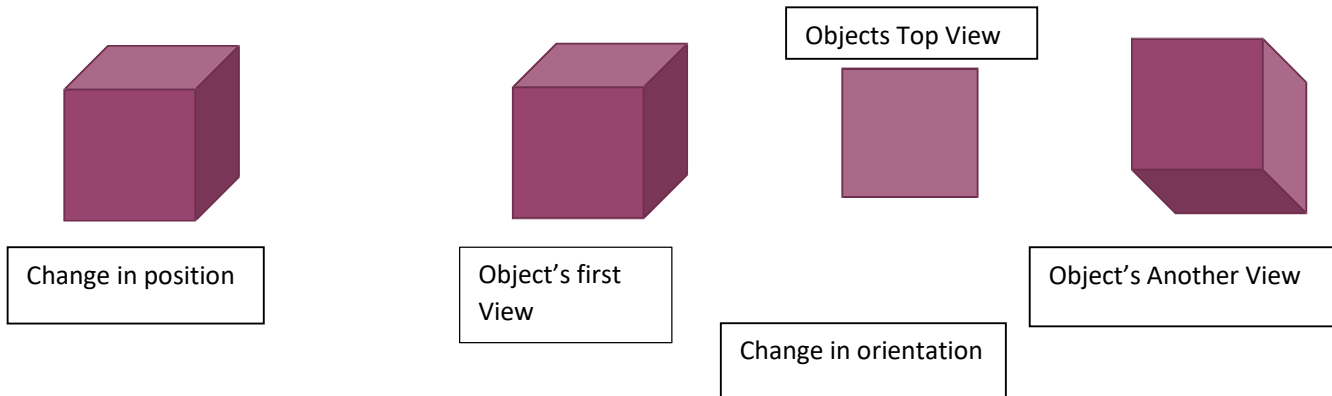
Some examples of haptic representation in VR:

1. **Haptic gloves:** These are gloves that use sensors and motors to simulate the sensation of touch and pressure on the user's fingers and palms. For example, a haptic glove might be used in a VR game to simulate the sensation of picking up and manipulating objects within the virtual world.
2. **Haptic vests:** These are vests that use motors to simulate the sensation of impact or pressure on the user's body. For example, a haptic vest might be used in a VR game to simulate the sensation of being hit or pushed by virtual objects or characters.
3. **Haptic controllers:** These are handheld controllers that use motors to create the sensation of vibration or force feedback. For example, a haptic controller might be used in a VR shooting game to simulate the sensation of firing a gun, or in a driving game to simulate the sensation of driving over rough terrain.
4. **Haptic floors:** These are floors that use motors or pneumatic devices to create the sensation of movement or vibration under the user's feet. For example, a haptic floor might be used in a VR rollercoaster simulation to simulate the sensation of movement and acceleration.

Haptic representation in VR is a rapidly growing field, with new devices and technologies being developed all the time. It has the potential to greatly enhance the user's sense of presence and engagement within the virtual environment, and can be used in a wide range of applications, from gaming and entertainment to education and training.

Geometric Models:

- A model can be any object in a Virtual World.
 - e.g. Chair, a tree, wall of room, user's avatar etc.
- Objects can be transformed
 - Change in position
 - Change in orientation



- A **virtual world** can be represented in 3D Euclidean space with cartesian coordinates
- Let R^3 represents a virtual world
 - Every point is represented as a triple of real-valued coordinates: (x, y, z)
 - We follow right-handed coordinate system

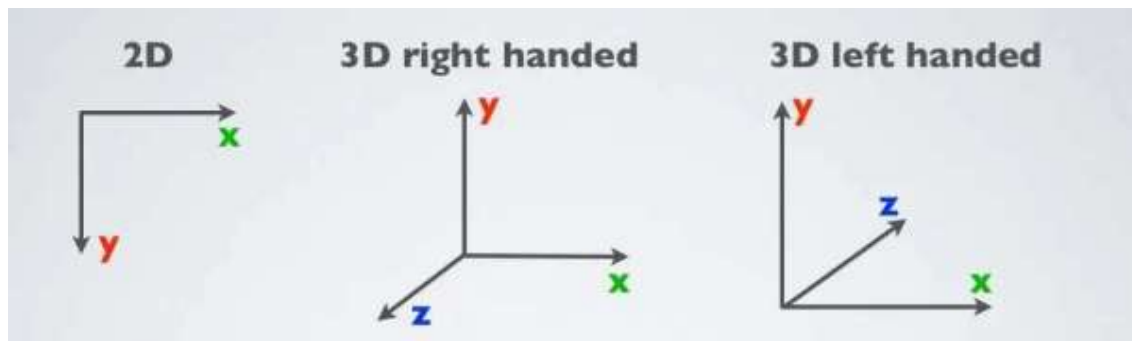


Fig. Co-ordinate System

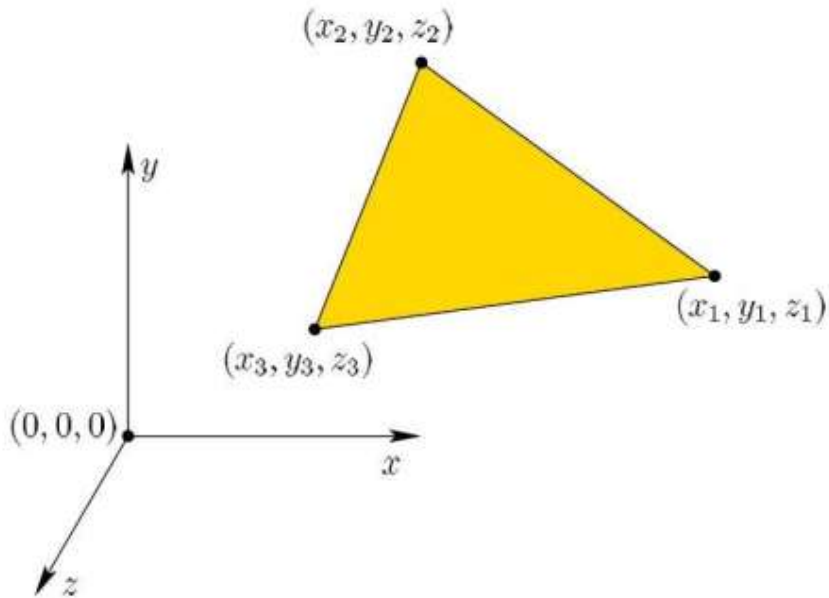
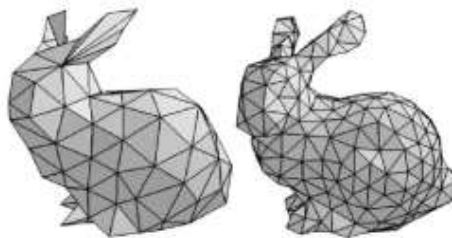


Fig.Co-ordinate axes of Virtual World

Geometric Models:

- Are made of surfaces or solid regions in R^3
- Contain infinite no. of points
- Are defined in terms of primitives
- Each primitive represents an infinite set of points
- e.g. 3D triangle
- Planar surface/ face of the triangle is specified by the coordinates of the triangle vertices $((x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3))$
- A complicated object or body in the virtual world modelled using numerous triangles can be arranged into a **mesh**.



Data Structures

- Choice of data structure for geometric models should depend on **which operation will be performed on the model**.
- Three kinds of data elements
 - **Face** represents *two*-dimensional part of the model
 - **Edge** represents *one*-dimensional part of the model
 - **Vertex** represents *zero*-dimensional part of the model
- The data structure contains ***pointers between adjacent faces, edges, and vertices*** so that algorithms can quickly traverse the model components.

Inside vs Outside

- The model forms a complete barrier between the inside and outside of the object.
- Coherent Model
- Use of high-end design tools like Blender/ Maya

Why triangles?

- The simplest for algorithms to handle
- Complicated primitives may lead to smaller model sizes but increases the cost of computation

Stationary vs Movable Models in Virtual World

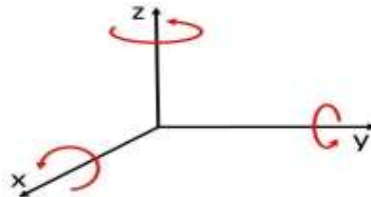
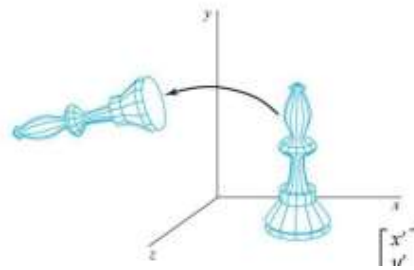
- ***Stationary models***, which keep the same co-ordinates forever.
- ***Movable models***, which can be transformed into various positions and orientations.
- Motion can be caused in following ways
 - Using tracking system
 - With the use of controllers
 - Self-movement according to the laws of physics

Changing position and Orientation:

Assume model is defined as a mesh of triangles.

Changing position: Translation

Changing orientation: Rotation



In virtual reality (VR), changing position and orientation in geometric models is a key feature that allows users to interact with and explore virtual environments. This involves the ability to move, rotate, and scale virtual objects within a 3D space, in order to create a more immersive and engaging virtual experience.

There are several ways that users can change their position and orientation within geometric models in VR, including:

1. **Teleportation:** This involves selecting a location within the virtual environment and instantly "teleporting" to that location, typically by pointing a controller or other input device at the desired location.
2. **Walking:** Some VR systems allow users to physically walk around within the virtual environment, either by using a treadmill or by tracking their physical movement within a designated space.
3. **Flying:** Some VR systems allow users to fly or hover within the virtual environment, either by using a controller or by tracking the user's hand gestures.
4. **Rotating:** Users can rotate their view of the virtual environment by using a controller or by physically turning their head, allowing them to view objects and environments from different angles.
5. **Scaling:** Users can scale virtual objects within the virtual environment, either by using a controller or by using hand gestures, allowing them to adjust the size and proportion of objects to suit their preferences.

Overall, the ability to change position and orientation within geometric models is a key feature of VR that allows users to explore and interact with virtual environments in a highly intuitive and immersive way.

Translation:

In virtual reality (VR), translation refers to the movement of objects or the user's point of view within the virtual environment. This involves moving objects or the user's position in three-dimensional space along one or more axes, such as up and down, left and right, or forward and backward.

Translation is an important feature of VR, as it allows users to move through virtual environments and interact with virtual objects in a more immersive and realistic way. For example, in a VR game or simulation, users may need to move around a virtual room or walk through a virtual landscape to complete tasks or solve puzzles.

There are several ways that translation can be implemented in VR, including:

1. **Handheld controllers:** Many VR systems come with handheld controllers that allow users to move their virtual avatar or manipulate virtual objects within the virtual environment.
2. **Treadmills:** Some VR systems use treadmills or other locomotion devices to allow users to physically walk or run through virtual environments.
3. **Teleportation:** Teleportation is a common method of translation in VR, where users can select a location within the virtual environment and instantly "teleport" to that location.
4. **Natural movement:** Some VR systems use sensors or other tracking technology to allow users to move their bodies in a natural way, such as walking or jumping, to control their movement within the virtual environment.

Overall, translation is a crucial feature of VR that allows users to fully immerse themselves in virtual environments and interact with virtual objects in a more intuitive and engaging way.

Rotation:

- Many times we need to change the mode *orientation* in the virtual world.
- The operation that changes the orientation is called *rotation*.
- Rotations in three dimensions are much more complicated than translations.

In virtual reality, rotation refers to the ability to move or turn your virtual viewpoint or perspective around an object, scene or environment. This movement can be achieved using various input devices such as handheld controllers, trackpads, or even by moving your head or body in physical space.

When you rotate in virtual reality, your viewpoint or perspective changes, allowing you to view the virtual environment from different angles. For example, if you are playing a virtual

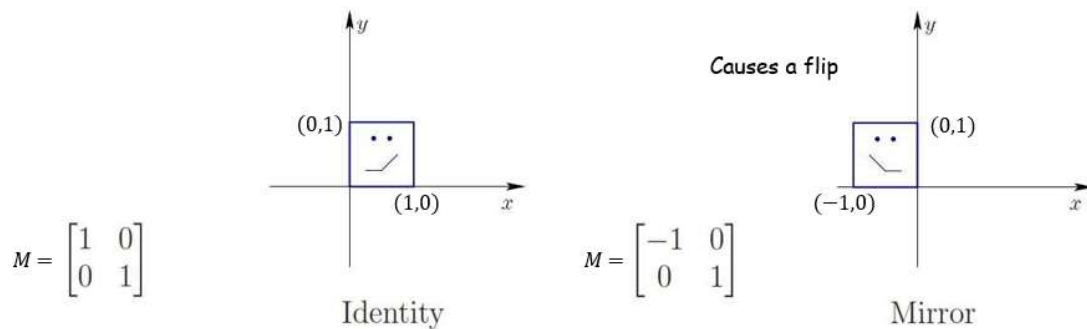
game and need to look around a corner or see something behind you, you can rotate your viewpoint in the virtual world to get a better look.

Rotating in virtual reality is often controlled by moving the input device or by using buttons or triggers to initiate the rotation. Some virtual reality systems also allow for natural rotation by tracking the movement of your head or body, allowing you to turn and look around in a more immersive way.

Overall, rotation is an important aspect of virtual reality that allows users to fully explore and interact with the virtual environment, providing a more immersive and realistic experience.

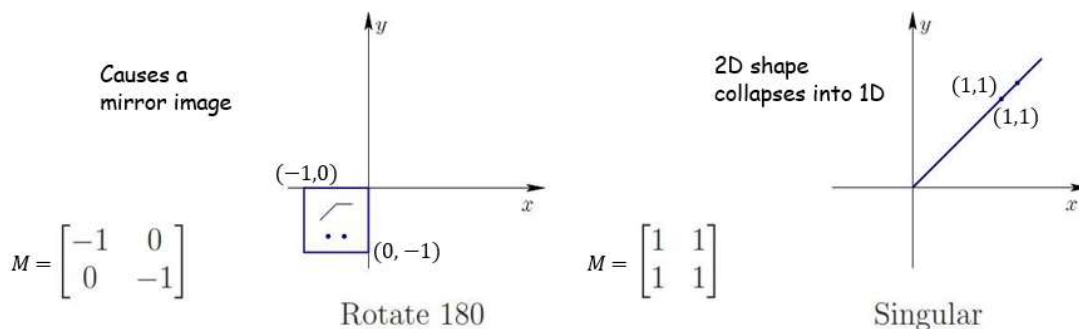
• 2D Linear Transformations

- Applying the 2D matrix to points



• 2D Linear Transformations

- Applying the 2D matrix to points



Which matrices produce rotations?

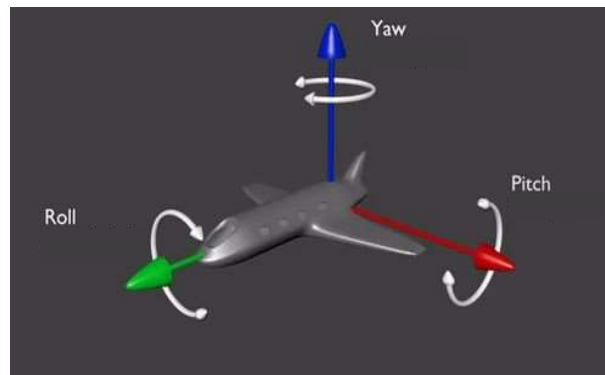
Matrix must satisfy the following rules to produce rotation:

1. No stretching of axes
2. No shearing
3. No mirror images

3D Case of Rotation

- There are 3 degrees of rotational freedom.
- So, it is possible to pick three independent parameters for 3D rotation and then derive all 9 components from them.
- Parameterizing 3D rotations by constructing them from “2D-like” transformations.

Yaw, Pitch and Roll



Yaw, pitch, and roll are three terms used to describe the movement and orientation of an object in three-dimensional space, such as an airplane or a spacecraft.

Yaw refers to the rotation of the object around a vertical axis. For example, if you were standing up straight and turned your head left and right without tilting it up or down, that would be a yaw motion.

Pitch refers to the rotation of the object around a horizontal axis that is perpendicular to the yaw axis. For example, if you were standing up straight and tilted your head forward and backward without turning it left or right, that would be a pitch motion.

Roll refers to the rotation of the object around a horizontal axis that is parallel to the ground. For example, if you were standing up straight and tilted your head to the side without turning it left or right or up or down, that would be a roll motion. Together, yaw, pitch, and roll allow us to describe the complete orientation and movement of an object in three-dimensional space.

Axis angle representations of Rotation:

In three-dimensional space, a rotation can be represented using an axis-angle representation. This representation describes a rotation in terms of a unit vector, known as the rotation axis, and an angle of rotation about that axis.

To represent a rotation using an axis-angle representation, we can use the following formula:

$$R = \exp(\theta * K)$$

Here, **R** is the rotation matrix corresponding to the rotation, **theta** is the angle of rotation, and **K** is the skew-symmetric matrix associated with the rotation axis. The skew-symmetric matrix **K** can be constructed using the following formula:

$$K = \begin{bmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{bmatrix}$$

Where **(kx, ky, kz)** is the unit vector representing the rotation axis.

The rotation matrix **R** can be computed using the matrix exponential function **exp()**:

$$R = \exp(\theta * K) = I + \sin(\theta) * K + (1 - \cos(\theta)) * K^2$$

Where **I** is the identity matrix and **K²** is the matrix product of **K** with itself.

The axis-angle representation of a rotation is useful because it provides an intuitive way to describe the orientation of an object in space. For example, a camera's orientation can be described using an axis-angle representation, with the rotation axis corresponding to the camera's pointing direction and the angle of rotation corresponding to the camera's field of view. Similarly, the orientation of a spacecraft can be described using an axis-angle representation, with the rotation axis corresponding to the spacecraft's axis of rotation and the angle of rotation corresponding to the amount of rotation.

Viewing transformations:

Viewing Transformation is the mapping of coordinates of points and lines that form the picture into appropriate coordinates on the display device.

World coordinate system (WCS) is the right-handed cartesian co-ordinate system where we define the picture to be displayed. A finite region in the WCS is called the *Window*. The corresponding coordinate system on the display device where the image of the picture is displayed is called the *physical coordinate system*. Mapping the *window* onto a subregion of the display device called the *viewport* is called the *Viewing Transformation*. *Normalized device coordinate (NDC)* is the display area

of the virtual display device corresponding to a unit square. The lower left-handed corner is the origin of the coordinate system. Mapping the window in the world coordinate space to viewport in NDC space is called the Normalization Transformation, N .

Viewing transformations are a set of mathematical operations used in computer graphics to transform the position and orientation of an object or scene in 3D space into a 2D image that can be displayed on a screen.

In order to display a 3D object on a 2D screen, viewing transformations are necessary to account for perspective, depth, and other visual effects that are lost in the projection from 3D to 2D space. These transformations involve translating, rotating, and scaling the object's position and orientation relative to the viewer's perspective.

Some common types of viewing transformations include:

- Translation: moving the object or camera position in the x, y, or z directions.
- Rotation: rotating the object or camera position around the x, y, or z axes.
- Scaling: changing the size of the object or scene.

Viewing transformations are important in many applications of computer graphics, including video games, virtual reality, and architectural and product design. By applying these transformations, designers can create realistic and immersive experiences for viewers, giving them the sense of being present in a 3D environment.

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